

# Ordinary Differential Equations Note

Differential Equations with Boundary-Value Problems, 9/e (Metric Edition)

## 2 First-Order Differential Equations

### 2.1 Solution Curves without a Solution

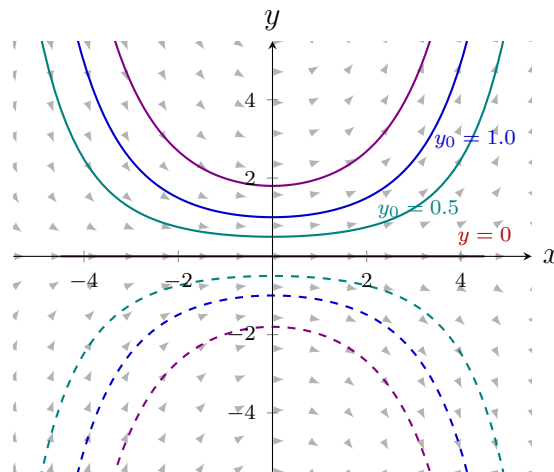
#### 2.1.1 Direction Field(方向場)

Find solutions with the direction field is a qualitative analysis(定性分析).

**Example.** The equation

$$y' = 0.2xy.$$

With the direction field and solution curve:



#### 2.1.2 Phase portrait for Autonomous ODE(自主性方程與相圖)

To solve the equation shown as the form

$$\frac{dy}{dx} = f(y),$$

we will use the following steps:

1. Equilibrium(平衡) / Critical Points: Let  $y' = f(y) = 0$  and find its critical points.

**Example.**

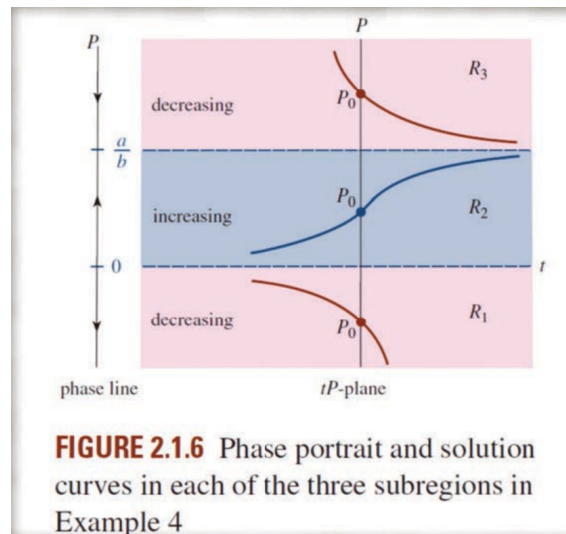
$$\frac{dp}{dt} = p(a - bp) = f(p).$$

Let  $p' = p(a - bp) = f(p) = 0$ , implies

$$p(a - bp) = 0.$$

Then  $p = 0$  or  $p = \frac{a}{b}$ .

## 2. Phase portrait



The **attractor**(吸引子) is a point that both arrows on either side of the phase line point to this point, which is the stable equilibrium point (e.g.  $p = \frac{a}{b}$ ); and the **repeller**(排斥子) is a point that the arrows on both sides of the phase line are moving away from the point, indicating an unstable equilibrium point. (e.g.  $p = 0$ )

## 2.2 Separable Equations

If  $\frac{dy}{dx} = g(x)$  and  $g(x)$  is continuous, then

$$\begin{aligned} \frac{dy}{dx} = g(x) &\implies \frac{dy}{dx}dx = g(x)dx \\ &\implies dy = g(x)dx \\ &\implies \int dy = \int g(x)dx + C \\ &\implies y = \int g(x)dx + C \end{aligned}$$

**Definition.** A first-order differential equation form

$$\frac{dy}{dx} = g(x)h(y)$$

or

$$\frac{dy}{dx} = \frac{g(x)}{h(y)}$$

or

$$\frac{dy}{dx} = \frac{h(y)}{g(x)}$$

is said to be **separable** or to have separable variables.

Case 1: If  $h(y) \neq 0$ <sup>1</sup>, then

$$\begin{aligned}\frac{dy}{dx} = g(x)h(y) &\implies \frac{1}{h(y)}dy = g(x)dx \\ &\implies \int \frac{1}{h(y)}dy = \int g(x)dx + C, C \in \mathbb{R} \\ &\implies \dots \\ &\implies y = \Phi(x) + \xi\end{aligned}$$

and

$$\begin{aligned}\frac{dy}{dx} = \frac{g(x)}{h(y)} &\implies h(y)dy = g(x)dx \\ &\implies \int h(y)dy = \int g(x)dx + C, C \in \mathbb{R}\end{aligned}$$

Case 2: If  $g(x) \neq 0$  and  $h(y) \neq 0$ , then

$$\begin{aligned}\frac{dy}{dx} = \frac{h(y)}{g(x)} &\implies \frac{1}{h(y)} = \frac{1}{g(x)}dx \\ &\implies \int \frac{1}{h(y)} = \int \frac{1}{g(x)}dx + C, C \in \mathbb{R}\end{aligned}$$

**Example.** Solve the equation

$$y' = \frac{y}{1+x}.$$

**Sol.**

$$\begin{aligned}y' = \frac{dy}{dx} = \frac{y}{1+x} &\implies \frac{1}{y}dy = \frac{1}{1+x}dx \\ &\implies \int \frac{1}{y}dy = \int \frac{1}{1+x}dx + C \\ &\implies \ln|y| = \ln|1+x| + C \\ &\implies e^{\ln|y|} = e^{\ln|1+x|+C} \\ &\implies y = e^C(1+x), C \in \mathbb{R}\end{aligned}$$

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<sup>1</sup>For the form  $\frac{dy}{dx} = g(x)h(y)$  and  $\frac{dy}{dx} = \frac{g(x)}{h(y)}$ .

**Example.** Solve the equation

$$y' = \frac{1}{1 + \sin x}.$$

**Sol.**

$$\begin{aligned} y' = \frac{dy}{dx} = \frac{1}{1 + \sin x} &\implies dy = \frac{1}{1 + \sin x} dx \\ &\implies \int dy = \int \frac{1}{1 + \sin x} dx + C \\ &\implies y = \int \frac{1}{1 + \sin x} \cdot \frac{1 - \sin x}{1 - \sin x} dx + C \\ &\implies y = \int \frac{1 - \sin x}{1 - \sin^2 x} dx + C \\ &\implies y = \int \frac{1 - \sin x}{\cos^2 x} dx + C \\ &\implies y = \int \frac{1}{\cos^2 x} dx - \int \frac{\sin x}{\cos^2 x} dx + C \\ &\implies y = \int \sec^2 x dx - \int \tan x \sec x dx + C \\ &\implies y = \tan x - \sec x + C, C \in \mathbb{R}. \end{aligned}$$

**Example.** Solve the equation

$$\frac{dy}{dx} = y^2 - 4.$$

**Sol.**

$$\begin{aligned} \frac{dy}{dx} = y^2 - 4 &\implies \int \frac{1}{y^2 - 4} dy = \int dx + C \\ &\implies \int \frac{1}{(y - 2)(y + 2)} = x + C \\ &\implies \int \frac{1/4}{y - 2} + \frac{-1/4}{y + 2} dy = x + C \\ &\implies \frac{1}{4} (\ln |y - 2| - \ln |y + 2|) = x + C. \end{aligned}$$

Thus

$$\begin{aligned} \frac{1}{4} (\ln |y - 2| - \ln |y + 2|) &= x + C \\ \implies \ln \left| \frac{y - 2}{y + 2} \right| &= 4x + 4C \\ \implies \frac{y - 2}{y + 2} &= e^{4x + 4C} \\ \implies y &= 2 \cdot \frac{e^{4x + 4C} + 1}{1 - e^{4x + 4C}}, C \in \mathbb{R}. \end{aligned}$$

So  $y = 2 \cdot \frac{e^{4x + 4C} + 1}{1 - e^{4x + 4C}}$  is a general solution of the equation.

Question: In the last example, is there any singular solution for the equation?  $y = 2$  and  $y = -2$  are solutions, but  $y = 2$  can be derived from the general solution. Hence  $y = -2$  is a singular solution.

**Example.** Solve

$$\cos x(e^{2y} - y)\frac{dy}{dx} = e^y \sin 2x$$

with  $y(0) = 0$ .

**Sol.**

$$\begin{aligned}\cos x(e^{2y} - y)\frac{dy}{dx} = e^y \sin 2x &\implies \cos x(e^{2y} - y)\frac{dy}{dx} = 2e^y \sin x \cos x \\ &\implies e^y - ye^{-y}dy = 2 \sin x dx \\ &\implies \int e^y - ye^{-y}dy = \int 2 \sin x dx + C \\ &\implies \int e^y dy - \int ye^{-y}dy = \int 2 \sin x dx + C\end{aligned}$$

Using integration by parts for  $\int ye^{-y}dy$ , this implies

$$\int ye^{-y}dy = -ye^{-y} - e^{-y}$$

Therefore,

$$\begin{aligned}\int e^y dy - \int ye^{-y}dy &= \int 2 \sin x dx + C \\ \implies e^y + ye^{-y} + e^{-y} &= -2 \cos x + C \\ \implies e^y + ye^{-y} + e^{-y} + 2 \cos x &= C, C \in \mathbb{R}\end{aligned}$$

Then  $e^y + ye^{-y} + e^{-y} + 2 \cos x = C$  is a general solution for constant  $C$ . Let  $y(0) = 0$ , thus we obtain

$$1 + 0 + 1 + 2 = C = 4.$$

Hence  $e^y + ye^{-y} + e^{-y} + 2 \cos x = 4$  is a solution with IVP.

## 2.5 Solutions by Substitutions

**Example.** Is  $2xyy' = y^2 - x^2$  separable? The answer is **NO!**

**Example.** From last example, we have

$$\begin{aligned} 2xy \frac{dy}{dx} = y^2 - x^2 &\implies \frac{dy}{dx} = \frac{y^2}{2xy} - \frac{x^2}{2xy} \\ &\implies \frac{dy}{dx} = \frac{y}{2x} - \frac{x}{2y} \end{aligned}$$

It can't solve by separable method, but if we let  $u = \frac{y}{x}$ , we obtain

$$\frac{dy}{dx} = \frac{y}{2x} - \frac{x}{2y} = \frac{1}{2} \left( u - \frac{1}{u} \right)$$

Since  $u = \frac{y}{x}$ , that is  $y = ux$ , then by chain rule and product rule,

$$\frac{dy}{dx} = \frac{du}{dx}x + u.$$

Therefore,

$$\begin{aligned} \frac{dy}{dx} = \frac{1}{2} \left( u - \frac{1}{u} \right) &\implies \frac{du}{dx}x + u = \frac{1}{2} \left( u - \frac{1}{u} \right) \\ &\implies x \frac{du}{dx} = \frac{-(1 + u^2)}{2u} \\ &\implies \frac{2u}{(1 + u^2)} du = \frac{1}{x} dx \\ &\implies \int \frac{2u}{(1 + u^2)} du = \int \frac{1}{x} dx + C \\ &\implies \ln(1 + u^2) = -\ln|x| + C \\ &\implies 1 + u^2 = \left| \frac{1}{x} \right| + e^C. \end{aligned}$$

Then take  $u = \frac{y}{x}$  back to the equation

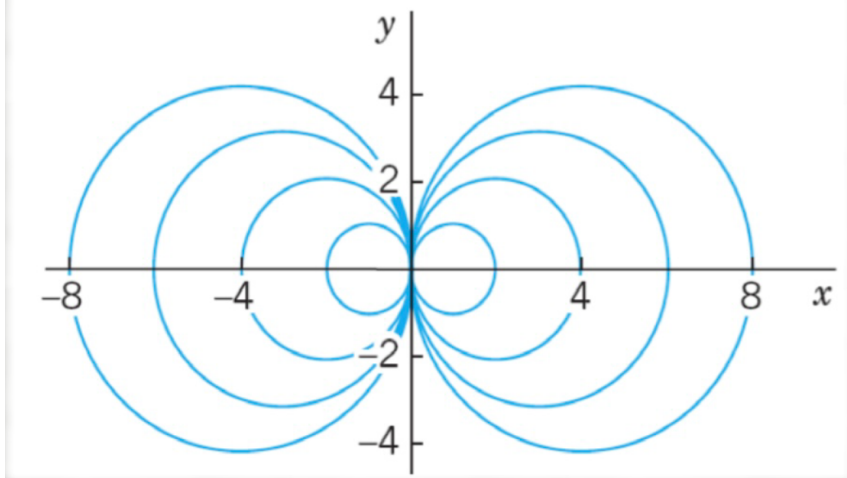
$$1 + \left( \frac{y}{x} \right)^2 = \frac{1}{x} + e^C \implies x^2 + y^2 = xe^C, \quad C \in \mathbb{R},$$

also we can let  $a = e^C$  then we obtain

$$x^2 + y^2 = xa \implies \left( x - \frac{a}{2} \right)^2 + y^2 = \frac{a^2}{4}.$$

Hence  $\left( x - \frac{a}{2} \right)^2 + y^2 = \frac{a^2}{4}$  is a general solution.

**General solution- family of circles (通解的函數圖形)**



Recall the chain rule for partial derivative on  $u = f(x, y) \implies y = h(x, u)$ :

$$dy = \frac{\partial y}{\partial x} dx + \frac{\partial y}{\partial u} du$$

**Example.** Solve

$$(x^2 + y^2)dx + (x^2 - xy)dy = 0.$$

**Sol.** Let  $u = \frac{y}{x}$ , then  $y = ux$ . Compute  $dy$ , we have

$$\begin{aligned} dy &= \frac{\partial y}{\partial x} dx + \frac{\partial y}{\partial u} du \\ &= u dx + x du \end{aligned}$$

Thus

$$\begin{aligned} &(x^2 + y^2)dx + (x^2 - xy)dy = 0 \\ \implies &(x^2 + u^2 x^2)dx + (x^2 - ux^2)(u dx + x du) = 0 \\ \implies &x^2(1 + u^2)dx + x^2(u - u^2)dx + x^3(1 - u)du = 0 \\ \implies &x^2(1 + u)dx + x^3(1 - u)du = 0 \\ \implies &\frac{1}{x}dx + \frac{1 - u}{1 + u}du = 0. \end{aligned}$$

Then

$$\begin{aligned}\frac{1-u}{1+u}du &= -\frac{1}{x}dx \implies \int \frac{1-u}{1+u}du = \int -\frac{1}{x}dx + C \\ &\implies \int \frac{1}{1+u}du - \int \frac{u}{1+u}du = \ln \left| \frac{1}{x} \right| + C \\ &\implies \ln |1+u| - (u - \ln |1+u|) = \ln \left| \frac{1}{x} \right| + C \\ &\implies 2 \ln |1+u| - u = \ln \left| \frac{1}{x} \right| + C \\ &\implies \ln(1+u)^2 - u - \ln \left| \frac{1}{x} \right| = C, \quad C \in \mathbb{R}.\end{aligned}$$

Hence  $\ln(1+u)^2 - u - \ln \left| \frac{1}{x} \right| = C$  is a general solution for constant  $C$ .